Case study: diagnosis, plan, proof design, trust appendix

Brad Sebastian | Prepared for TriFlow Solutions | 2 September 2026

Workings are in the attached workbook; every figure here is a cell in it, with assumptions labeled A1 to A15.

1. Diagnosis

Three-quarters of revenue comes from customers the company already has and the crews are full, so new Restoration demand goes to subcontractors at 15% margin. The plan grows the 64%-margin Maintenance line first, claims $901K of year-one revenue almost all from customers and relationships the company already has, and puts the first new customer I can fairly credit to marketing at month eighteen.

Where growth actually comes from

The customers we already have are where the money comes from. They bought all of the Maintenance and licensing work, the equipment line and 30 of the 58 Restoration jobs last year, which is about $6.35M of the $8.4M, or 76%. The new customers came in almost entirely through people who already knew us: of the $2.05M in new-customer Restoration revenue, $1.62M (79%) was 18 wins that started as a customer referral or an introduction from a consulting engineer. I am reading Exhibit 6's $2.05M as all new-customer Restoration revenue, which its job count supports and its wording does not say; the true figure is probably a little lower because some equipment work rides along. Those two sources closed half of what they touched (18 wins on 36 inquiries) at zero media cost, and nobody owns them; whoever picks up the phone handles it.

The paid channels are the opposite picture: paid search and the trade show cost $75K, produced 4 wins and $134K of revenue, and at Restoration's 27% margin returned about $36K of gross profit, a loss of roughly $39K before anyone's salary. Paid search alone took 210 inquiries to produce 2 jobs.

The real constraint

The company has more work than crews. It never turns down a job, it schedules six weeks out on a whiteboard, it has one licensed lead in its tightest state, and its subcontractor bill went from $22K to $840K in two years. Exhibit 5 says the same: 58 jobs at 2.6 crew-weeks and 424 visits at 0.35 is 299 crew-weeks of work, and the crews had 220. The gap closes if the $1.64M of subcontracted revenue is Restoration work (A1: about 30 jobs at $54K each, inside the municipal range): our own crews then delivered 28 jobs and every Maintenance visit, 220 crew-weeks. The tidy 100% is partly my assumption coming back to me: if the subcontracted jobs are the overflow, what is left is 220 by definition, and nothing in the pack can check it. I asked Amanda whether the subcontracted work is Restoration, and she confirmed it, so A1 is no longer only mine; a month of timesheets would settle the rest.

This is where the margin went. Over two years revenue grew $1.03M and subcontracted revenue went from near zero to $1.64M. All the growth was bought, and then some: our own crews delivered less than two years ago (appendix). Self-performed Restoration earns about $28K of gross profit per job, a subcontracted job about $8K, and the $20K difference across the 30-odd subcontracted jobs is roughly $610K a year, about 40% of adjusted EBITDA. Licensing a new crew lead takes five years in its tightest state, so until a sixth crew exists where it can be licensed, more Restoration demand from marketing produces revenue at 15% margin, delivered by someone else. Nobody hires a marketing director to fix crew capacity, but every number in my plan runs into it.

The recurring line is leaking

Maintenance is where the profit is, 49% of gross profit on 30% of revenue, and last year it lost more assets than it gained: 7 enrolled, 9 lost. Revenue still grew, mostly on the escalator (appendix). The board plans on a 34% conversion rate; last year 7 assets were enrolled against 58 jobs, and with a 14-month typical wait between restoration and contract, neither figure says how many restored customers convert. Of the 424 assets on maintenance, 176 are on a handshake with no contract and no escalator (about $1.05M a year at the average rate), which forgoes about $52K a year in escalator and compounds. At least 41 agreements come due inside twelve months with no renewal process, and 41 is a floor because two-thirds of agreement records have no expiry date. Meanwhile 480 restored assets at active customers receive no maintenance, and twenty years of before-and-after performance data has never been shown to a customer.

Two things the math turned up that I would want the team to look at

- A crew-week is worth about the same in either line if A1 holds, $10.8K of gross profit on self-performed Restoration and $10.9K on Maintenance. A crew-week on Maintenance gives up nothing that week, and the customer keeps paying for five contracted years, where a Restoration job pays once.

- Every email open and click figure in the pack is contaminated: a 71% government open rate with every link clicked within seconds is security software. I would not manage anything by those numbers; re-running last quarter's click report without clicks inside ten seconds of delivery should take them near zero.

The numbers I would manage this business by: gross profit per crew-week, with the share of Restoration revenue subcontracted beside it; Maintenance assets under signed agreement and net enrollments per quarter; and introductions per quarter from customers and consulting engineers, which nobody counts today.

2. The plan, with the arithmetic

The plan follows the order on the workbook's Quarters sheet: whatever turns into gross profit soonest goes first, whatever uses no crew time goes first, and nothing that brings in more Restoration work starts until inbound inquiries have a named owner and a promised response time and quotes go into a register. That puts the handshake conversion first even though it carries the smallest year-one gross profit of any revenue-bearing motion ($31K against the cross-sell's $137K), because the ask is a signature, it uses no crew-weeks, and the $1.05M of revenue behind it can walk at any time, which makes it the biggest number in the plan. Year-one revenue in the table comes from customers and relationships that already exist, apart from two awards the engineer firms would introduce, and the one program aimed at customers who have never heard of us claims no revenue this year.

Motion	Start	Year-one revenue	Year-one gross profit	Run-rate gross profit	Own crew-weeks	Spend
1a. Convert 176 handshake assets (A2: 60%)	Q1	$31K	$31K	$31K	0	$10K
1b. Renew the 41 known expiries (A5: 85% retained); protected, excluded from totals	Q1	$207K (excluded)	$133K (excluded)		0	$5K
1c. Cross-sell Maintenance to the 480 un-enrolled assets (A3: 15% enroll; A4: half bill in year)	Q1	$214K	$137K	$274K	25	$40K
2. Engineer program: referral ask at close-out, 8 named firms (A6: +4 introductions; A7: 2 awarded)	Q1	$258K	$39K	$39K	0	$30K
3. Multi-site utility parents: portfolio proposals for the president (A8 to A10: 1 award of 5 systems)	Q2	$225K	$34K	$10K	0.9	$10K
4. Lapsed outreach to 52 (A11, A12: 5 back, 3 jobs)	Q1	$173K	$26K		0	$5K
5. Deficiency outbound to the 190 top-tier fits, two waves (A13: zero revenue claimed)	Q2	0	0	year two	0	$35K
6. Foundations: CRM asset records, source field, quote register, routed inbound, website repair	Q1	0	0		0	$45K
7. Paid pilot, industrial/remediation, if 6 is live	Q3	0	0		0	$15K
Total, plus $5K measurement and contingency (spend includes 1b)		**$901K**	**$267K**	**$353K**	**26**	**$200K**

New Restoration work is taken at the 15% subcontractor margin throughout, because the crews are full under A1 and a sixth crew is finance's decision, not marketing's. Before displacement, year one returns about $267K of gross profit against $200K of spend (the escalator revenue in 1a is pure price on work already delivered, so nearly all of it is gross profit). After it, year one is about $34K negative: the new Maintenance visits take about 26 crew-weeks, which pushes about 10 Restoration jobs to subcontractors, half of them inside year one. From year two the program is worth about $152K a year net of displacement, about $137K after the $15.5K of foundations cost that recurs. The 72 assets in motion 1c secure about $1.37M of gross profit over their five-year terms. I will be candid that the 15% is a target I set; nothing in the pack derives it, and the workbook runs it from 5% to 25%. Customers added 7 assets on their own last year, under 2% of the 480, so it is the softest number in the plan and the week-eight and month-six checks sit under it.

Quarters one and two

Q1. Week one goes to getting the records straight (section 3). Weeks two through thirteen: the salesperson takes the 68 handshake customers with a signature ask and the escalator, the renewal calendar goes live with an expiry date on all 248 agreement records, and cross-sell wave one goes to half of the accounts holding the 480 assets, with the asset's own before-and-after performance record as the reason to call. The referral ask becomes a step in every project close-out, and the eight consulting-engineer firms become named accounts with a package built from the 102 assets they already brought us. Lapsed outreach wave one goes to 26 of the 52. Paid search and the trade show stop. That is about 175 conversations for one salesperson in 12 selling weeks, fewer in practice because many are the same customer, and marketing prepares the asset record and the offer for every call. I take the four multi-site parents with the president and the eight engineering firms personally, so the selling does not all land on him. The risk in the signature ask is that a customer who buys through procurement treats it as a tender; the week-eight rule catches that before the ask reaches the rest of the 68.

Q2. Outbound wave one goes to half of the 190 top-tier fits, matched to the other half on state, size and deficiency severity (section 3). The president gets a portfolio map of every system each of the four multi-site parents owns, flagged for deficiencies, and the website's two broken links are fixed and a routed inquiry form goes live. **Q3, in sketch.** Cross-sell wave two, lapsed wave two, and the month-six rule on the cross-sell. A small paid pilot for industrial and remediation buyers only (they spend from an appropriated budget and can move in weeks) under a rule agreed in advance: at $10K spent, stop if cost per site visit is above $3K, and the last $5K is released only if it passes. At month nine the outbound rule is called. **Q4.** Outbound wave two goes to the second half of the 190, untouched for two quarters; the renewal cycle runs on a complete record for the first time, the month-twelve rule on the engineer program is called, and the sixth-crew case goes to finance. Year two gets planned from the comparisons, not the 34%. What I did not get to in the six hours I spent on this: a quarter-by-quarter phasing of the table above and a by-state view of the 190, both week-two work.

What the cross-sell costs in crew-weeks, and the crew question

Those 26 crew-weeks (the 72 cross-sell assets plus the portfolio systems that enroll, at A16) give up about $201K of Restoration gross profit on the 10 displaced jobs against $283K of Maintenance gross profit gained, net about $82K a year before the escalator. A sixth crew is 44 crew-weeks: about $339K a year of gross profit if it brings subcontracted Restoration back in-house, or up to about $478K if it serves Maintenance, which needs 126 enrolled assets this plan does not yet produce; on the assets this plan does enroll, a sixth crew is worth the $201K of displacement it avoids, before the cost of the crew, which is not in the pack (job-level margin already carries crew labor on delivered work, so the real cost to net out is equipment, idle weeks and the licensing path). That decision is not mine, but it is the most valuable one in the plan and I would ask to model it with finance in Q1.

Where the money is not going, and why

- **Paid search goes to zero in the first half.** It lost money on gross profit, its leads had nowhere to land and nobody answered them, and its reported conversion mixes this year's inquiries with awards from year-old contacts.
- **No trade-show booth.** $41K for 2 wins. If the eight engineering firms are at a show, I would rather buy them lunch.
- **No new-logo Restoration demand generation in the first half.** Every job would be delivered at 15%.
- **No enterprise marketing automation and no website rebuild.** At 900 company records, 1,140 assets and 447 inquiries a year, a mid-tier CRM with custom objects is the right size; I have built exactly that and would rather spend on the data than on the platform. The $45K foundations line is roughly $12K a year for the CRM tier, $15K of implementation help, $10K to fix the site's two broken links, add a routed form and move it to hosting that stays up, $5K to load the 1,140 asset records, and $3K for call routing; about $15.5K of that recurs in year two. The $35K on motion 5 buys list data and enrichment for the 190, a contractor to build and load the matched pairs, and the outreach itself. No agency retainer: contractors for the data packaging and the CRM build, then done.
- **The salesperson stops cold outbound.** Last year it produced 41 inquiries, 5 site visits and 2 jobs. The same hours pointed at the 68 handshake customers, the 41 renewals and the 480 assets produce more, sooner, at 64% margin. The president keeps the four parents, and I work them with him.

If nothing closes for a quarter, the Restoration awards in motions 2, 3 and 4 slip, and that is $656K of the $901K. What slips least is the handshake conversion, about $31K of gross profit in year one and, at the 60% in A2, about $630K a year of the $1.05M handshake revenue put under contract for five years, because those customers already pay us and the ask is a signature. The cross-sell, about $137K, would slip too; those enrollments go through procurement, which is why the gate measures enrollment and procurement status instead of billings.

3. Proof design

Every system the company could sell to can be listed by name, so I need no attribution model. I can offer some customers and hold others back, compare the two, agree with the CEO in advance what result moves the budget, and say what a nineteen-month cycle leaves unknown.

What can be proven

The cross-sell. The accounts holding the 480 assets are split at random into a contact-now half and a six-month-wait half (by account, since the conversation is with the customer). Both halves get the same offer six months apart, and the month-six comparison is assets enrolled, contacted against waiting, with assets still in procurement reported separately (week eight counts enrolled or in procurement, since procurement rarely finishes in eight weeks). The waiting group's expected rate is close to the organic one: 7 assets added last year across the whole base, under 2% of the 480 addressable. One complication: many of the 68 handshake customers also hold un-enrolled assets, so the Q1 signature ask goes to both halves and only the cross-sell offer waits; a waiting-half customer who raises un-enrolled assets unprompted is logged and reported separately. Holding half back has a cost: at the 15% assumption about 36 of the 72 enrollments are in the waiting half and arrive two quarters later, roughly $107K of revenue delayed, not lost (72 assets, $5,943 an asset-year, half billing in year). The test is small, and I want the CEO to know that before he sees a result. If the 480 assets sit with about 80 accounts (A14, asked), counting by asset can see a gap of about 7 points and counting by account only about 17; since the split is by account the truth sits between. So the rule is set by asset, where 8 points can be seen, with the gap by account beside it. Assets clearing 8 with nothing per account buys one more quarter, not a pass. Both calculations are in the workbook.

Outbound to the 190. Matched pairs on state, system size and deficiency severity; one of each pair contacted in Q2, the other not before Q4. Nobody has ever contacted these systems, so the untouched group is a baseline for two quarters, and the rule is called at month nine, before wave two starts. Before the split, the 190 are checked against the 112 active customers and the four multi-site parents; any overlap goes to the president's list, outside the test. Measured on first meetings, site visits and registered quotes through year two, when awards land. Registered quotes are the result that matters (meetings mostly show that calling people produces meetings), so the month-nine rule includes them and revenue waits for year two. **Lapsed customers** get the same two waves.

What can only be inferred

The engineer program: introductions per quarter against last year's 36 from customers and engineers, by firm; eight firms is too few for a comparison group, so this is a trend the source and "referred by" fields at least make visible. Renewal rate: there is no baseline today, so year one becomes the baseline. First-touch source on every new record: real information, self-reported, and a long way from causation.

What is unknowable inside year one, and what I would say to the CEO about it

A historical win rate. Nobody kept a quote register, and in six years not one loss was recorded, so there is nothing to divide the wins by. Which of last year's 28 wins marketing touched: 11 have no source, and every loss was deleted. Anything from the email statistics. And, inside year one, new-customer Restoration revenue credited to marketing, a cost per acquisition on anything with a nineteen-month cycle, or a return on the $200K; 22 of last year's 28 wins had their first contact more than a year before award. New-customer revenue marketing can fairly claim arrives around month eighteen, counted from first contact, and I will say so in week one, not at the gate.

Week one

Freeze the four lists (480, 176, 41, 52) as a dated baseline. Load all 1,140 assets as records with restore date, maintenance status, contract or handshake, expiry and whether the escalator was applied. Make source mandatory with a closed list, add "referred by", add a loss stage with a reason and stop deleting. Write the definition of qualified (a named system, a named decision-maker, a stated problem and a budget source) into the CRM as a checklist, and start the quote register, in a shared sheet if the CRM cannot do it by Friday. Route every inbound inquiry to a named owner with a timestamp, drop email opens and clicks for replies and meetings, and put committed versus available crew-weeks, by region, into a sheet finance can see.

Decision rules, agreed before the results exist

When	Measure	Rule agreed in advance	If it misses
Week 8	The 480, 176 and 41 loaded with an owner and a contact	100%	In my control, so a miss is on me.

When	Measure	Rule agreed in advance	If it misses
Week 8	New records with a source recorded	95% or more	Fix the form.
Week 8	Median inbound response time	Under one business day	Fix routing before any demand spend.
Week 8	Handshake assets under signed agreement	20 or more (11% of 176)	Fewer than 10: the offer is wrong, probably the escalator. Fix terms.
Week 8	Cross-sell assets enrolled or in procurement, wave one	12 or more (5% of the roughly 240 assets in wave one; 2.5% of the 480)	Fewer than 6: pause and diagnose (8 weeks against a 14-month median).
Week 8	New-customer revenue from marketing	Not reported	Cannot be known yet; page says so.
Month 6	Cross-sell assets enrolled per asset, contacted vs waiting; in-procurement and account-level gap beside it	8 points or more (the test sees about 7): year-two budget and sixth-crew case go behind it. Under 3: budget moves to motions 2 and 5	3 to 8, or 8 per asset with nothing per account: one more quarter, wait group still waiting. Waiting half gets the offer either way.
Month 6	Lapsed reactivated, contacted vs waiting	3 or more against 0 or 1	Otherwise stop and leave them be.
Month 9	Outbound meetings, visits, quotes: wave one vs wave two, untouched until Q4	15 meetings, 5 site visits and 3 registered quotes against 2 or fewer of any	Otherwise stop; year two goes to engineers and parents.
Month 12	Customer and engineer introductions	40 or more against last year's 36	Trend only; too few firms to compare.
Month 12	Paid pilot, industrial and remediation	Cost per site visit under $3K at $10K spent	Stop at $10K; last $5K unspent.
Month 18	Attributed new-customer revenue	First honest report, by cohort	Anything earlier would be a guess.

4. Trust appendix: figures I would not bet money on yet

- **28 new customers (Exhibit 6).** At that pace there would be about 100 new Restoration customers since January 2023 against Exhibit 4's 76, and Exhibit 2 counts 164 customers in eight years, only 10 of whom bought exactly once; if 28 were new last year, at least 18 already bought a second time inside twelve months, on a nineteen-month cycle. Check: what "new customer" means in the CRM, then rebuild from invoices.
- **112 + 52 = 164 (Exhibit 2).** Every customer in eight years was active inside the last four, which cannot sit next to Exhibit 3's 236 assets "at a customer long gone." Either the 52 is everyone who ever lapsed, or the 164 is not eight years. Check: first and last invoice date by customer; it decides how clean the motion 4 list is.
- **299 crew-weeks against 220 (Exhibit 5).** Amanda confirmed the subcontracted $1.64M is Restoration work, the reading the reconciliation needs; if crews also service equipment, they are past 100%. If subcontracting went from near zero to $1.64M and revenue grew $1.03M, own crews delivered about $500K less than two years ago. Check: a month of timesheets.
- **"Top decile" (Exhibit 8).** 190 is 0.15% of 128,000, 1% of the 19,400 licensed-state systems and 4.4% of the 4,290 scored. Top decile is the wrong label, and the 5.9x lift was calibrated on known customers, which is circular. Check: rescore the 12,650 deficiency systems; do the 190 rise to the top?
- **$205K average lifetime revenue (Exhibit 2).** 164 customers times $205K is $33.6M; eight years of revenue at anything like today's growth rate is $45M to $55M before licensing. Either "lifetime" is narrower than it sounds or the figure is stale. Check: lifetime revenue by customer from accounting.
- **41 renewals due (Exhibit 3).** From the third of records with an expiry date; scaled it could be 120, and averaged over 248 five-year agreements about 50. Check: populate every agreement's expiry date.
- **34% conversion (Exhibit 4).** A customer-level figure since 2023 that the board plans on; last year 7 assets were enrolled against 58 jobs, and the program lost assets net. Check: enrollment by restoration cohort, and whether the 26 converters cluster in one segment.
- **The escalator.** Maintenance revenue grew about 7% over two years (Exhibit 1 note) on a base that lost assets net, roughly what 5% a year on the 248 contracted assets would give, so either it was applied and the 3% account is an exception, or something else grew. Check: the escalator, invoice by invoice, on all 248.

I would check the crew-weeks and the new-customer count first, with the CEO in the room, because the plan rests on both.

Method and AI use. Claude is my primary AI engine. I applied my drafts on this case to the work engine I have built inside it. I produced the memo and the workbook; the engine enhanced them: it tabulated the exhibits, rechecked the arithmetic, and its agents proofed the drafts for errors and consistency over multiple cycles. A catalog of the rules, skills, gates and workflows I have built inside Claude is at **bradsebastian.com/agentic**.